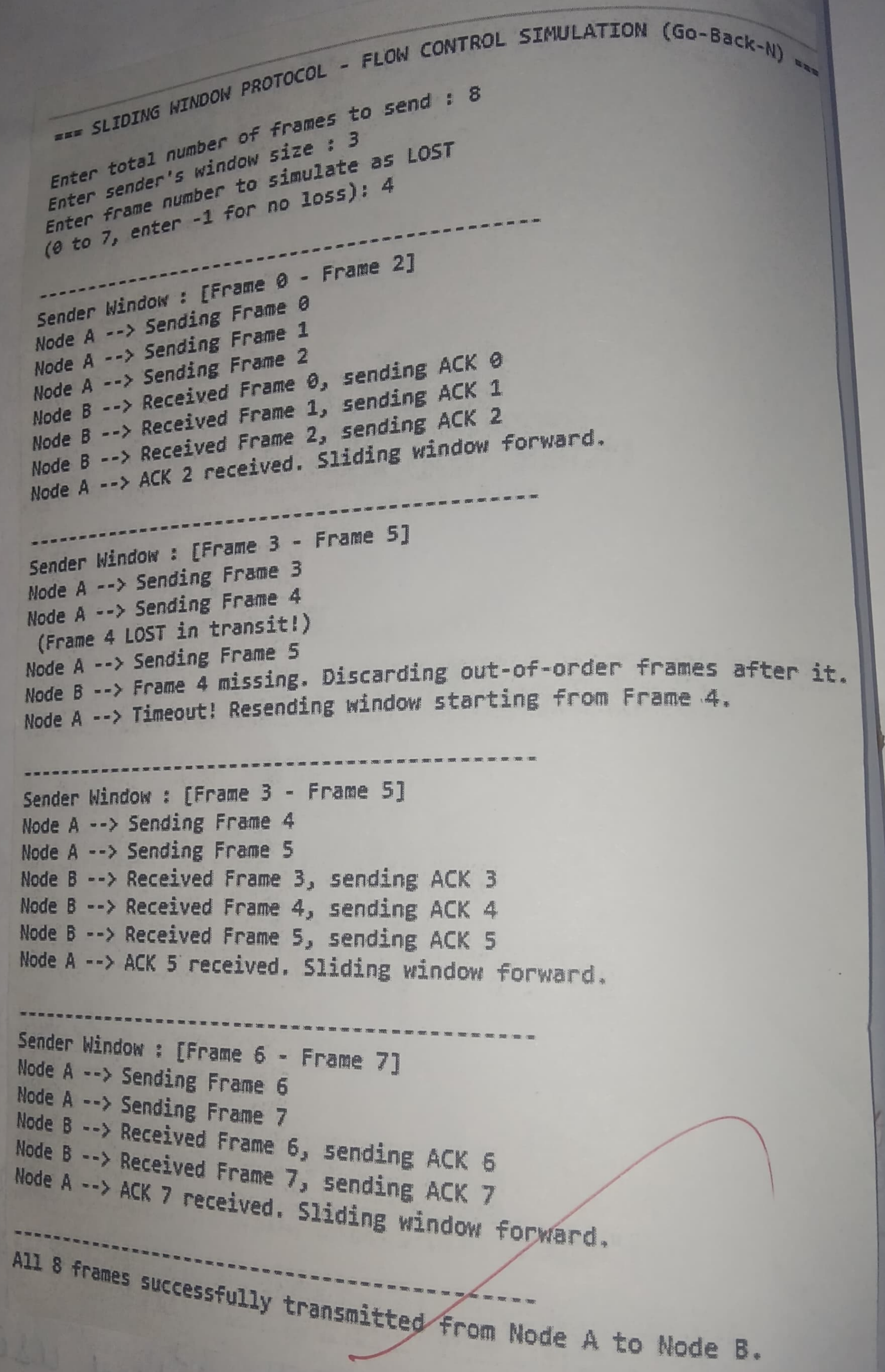
Write a program to implement flow Control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

STEPS:

1. Input Window size from the user
2. Input a text message from the user
3. Consider 1 character per frame
4. Create a frame with following fields [Frame no., DATA].
5. Send the frames.
6. Wait for the acknowledgement from the reciever.
7. Reader a file Called Receiver_Buffer
8. Check ACK field for the Acknowledgement number
9. If the Acknowledgement number is as expected, send new set of frames accordingly, Else if NACK is recieved, resend the frames



```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send : 8
Enter sender's window size : 3
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss): 4
----------------------------------------
Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.
----------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
 (Frame 4 LOST in transit!)
Node A --> Sending Frame 5
Node B --> Frame 4 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 4.

----------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node A --> ACK 5 received. Sliding window forward.

----------------------------------------
Sender Window : [Frame 6 - Frame 7]
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.
----------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
```

Create a receiver file with following features

1. Reader a file called sender_Buffer
2. Check the frame no.
3. If the frame sent, the Act for the window forward base = real seq
4. Show the program.

PROGRAMS

```
def simulate(total_frames, windows_size
 last base=0)
 next seq=0
while base < total frames:
 window_end=min(base+windows_size-
 total_frames-1)
 Print("\n_______________"
 Print(f"Sender window: [frame{base} frame
 {window_end}]")
 While result_seq<base+windows_size and
  next_seq <total_frames)
 Print(f"Node A → Sending frame (sent
 neat_seq±=1
 if lost_frame is not None and base
  <= lost_frame <next_seq:
 Print(f"Node B → frame {lost frame}
 missing Descending order
```

OUTPUT

```
=== SLIDING WINDOW PROTOCAl_ flow
CONTROL SIMULATION (GO_Back)===
link tot. no. of frame : 8
link Senders window size : 4
link frame no to simulate as
(0 to 7, enter -1, for loss): 5

Sender window: [frame 0 - frame 3]
Node A → Sending frame 0
Node A → Sending frame 2
Node A → Sending frame 3
Node B → Recieved frame 0, Send 0
Node B → recieved frame 2, Send 1
Node B → recieved frame 3, Send 3
Sender window: [frame 4 - frame 7]
Node A → Send frame 5
Node A → Send frame 6
Node A → send frame 7
Node B → Recieved frame 4, Send 4
Node B → Recieved frame 5, Send 5
Node B → recieved frame 7, Send 7
forward
- - - - - - - -
All 8 frames Successfully Transmit
from node A to node B
```

RESULT:

the program to implement flow control at the data link layer using sliding windows Protocol was successfully written in python, executed and verified by simulating the flow of frames b/w the networks